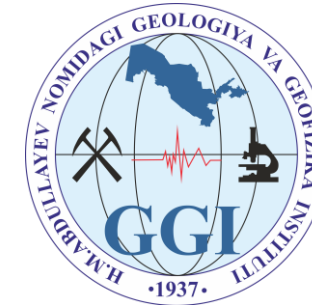


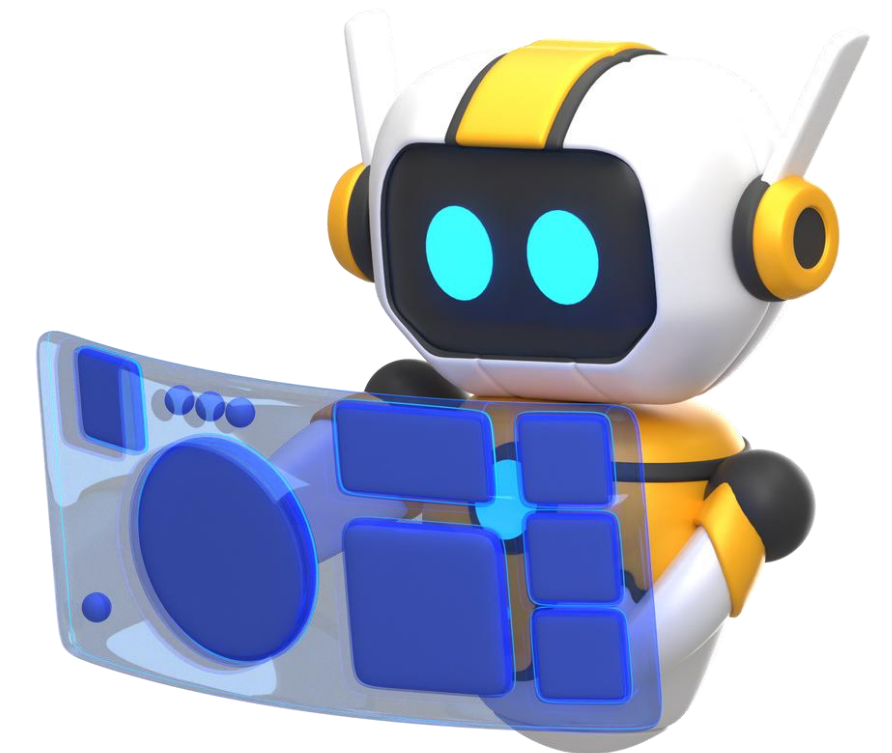


**TOG‘-KON SANOATI VA GEOLOGIYA VAZIRLIGI**

**“H.M. ABDULLAYEV NOMIDAGI GEOLOGIYA VA GEOFIZIKA INSTITUTI” DM**

**“PRESIDENT AI AWARD” tanlovi uchun**

**PETROLOG AI — Sun’iy intellekt asosida  
mineralogik-petrografik tahlil platformasi**



**YO‘LDOSHEV ZOKIR DONIYOR O‘G‘LI**

# LOYIHA ISHTIROKCHILARI



YO'LDOSHEV ZOKIR  
DONIYOR O'G'LI

**Kichik ilmiy xodim.**  
**Loyiha rahbari**



**Yoshuzoqov Shohboz  
G'aybulloyevich**

Shlif ma'lumotlari  
tahlilchisi



**Keldibekova Zilola  
Asror qizi**

Loyiha doirasida  
bajariladigan kameral  
va laboratoriya ishlari  
uchun ma'sul

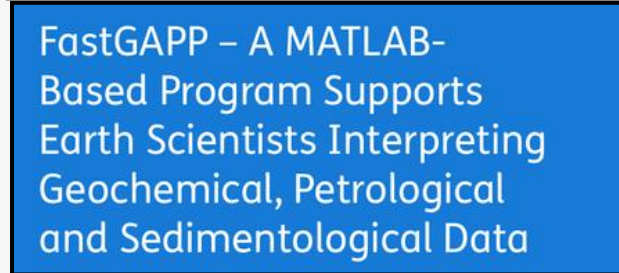
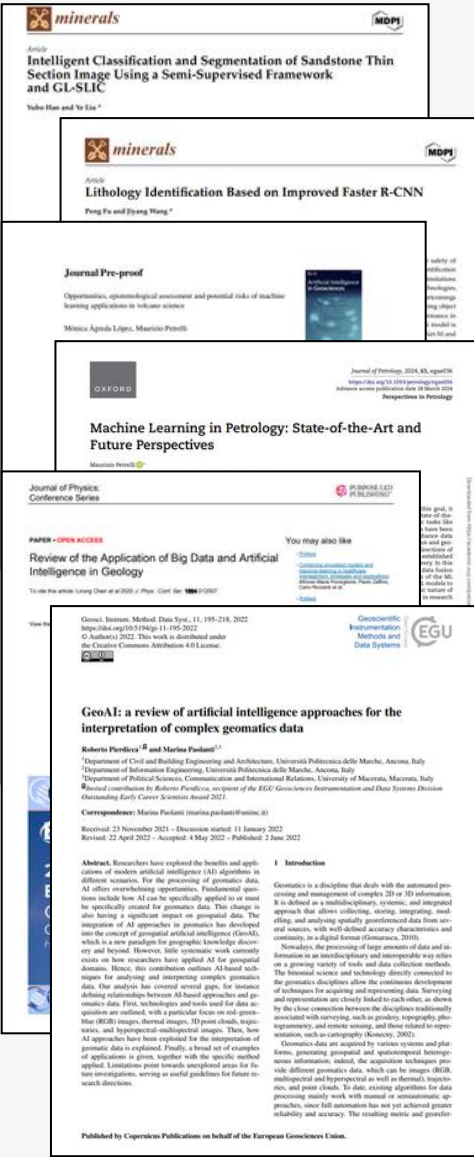
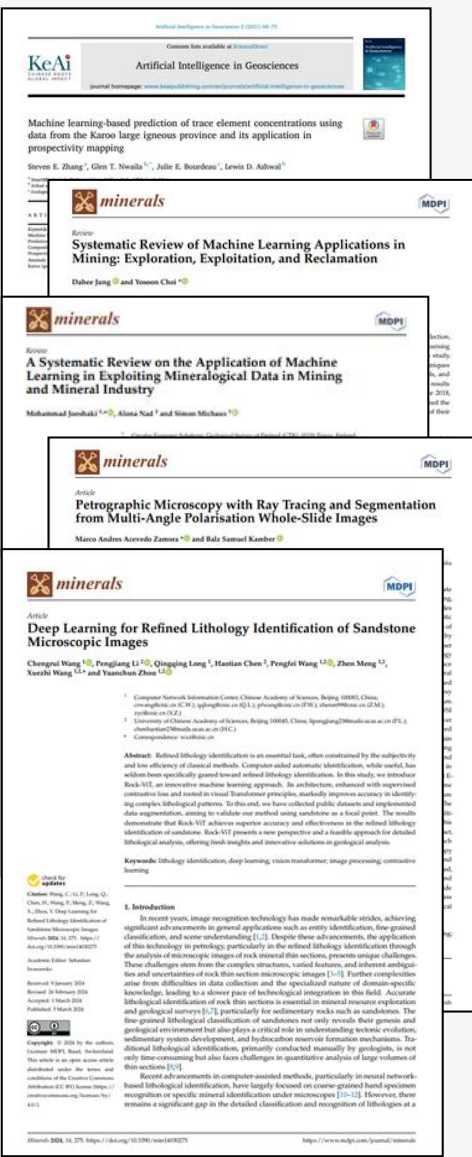
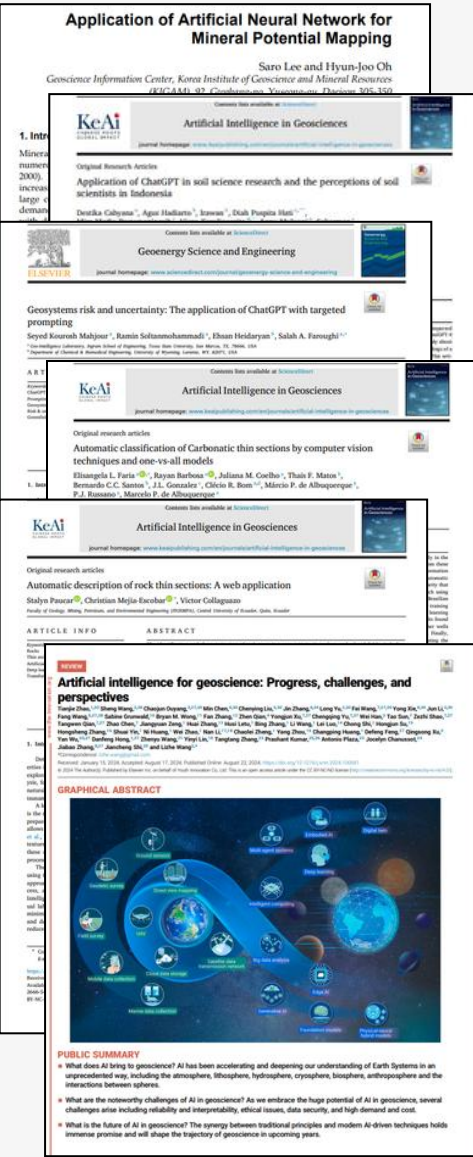


# MUAMMONING O‘RGANILGANLIK DARAJASI

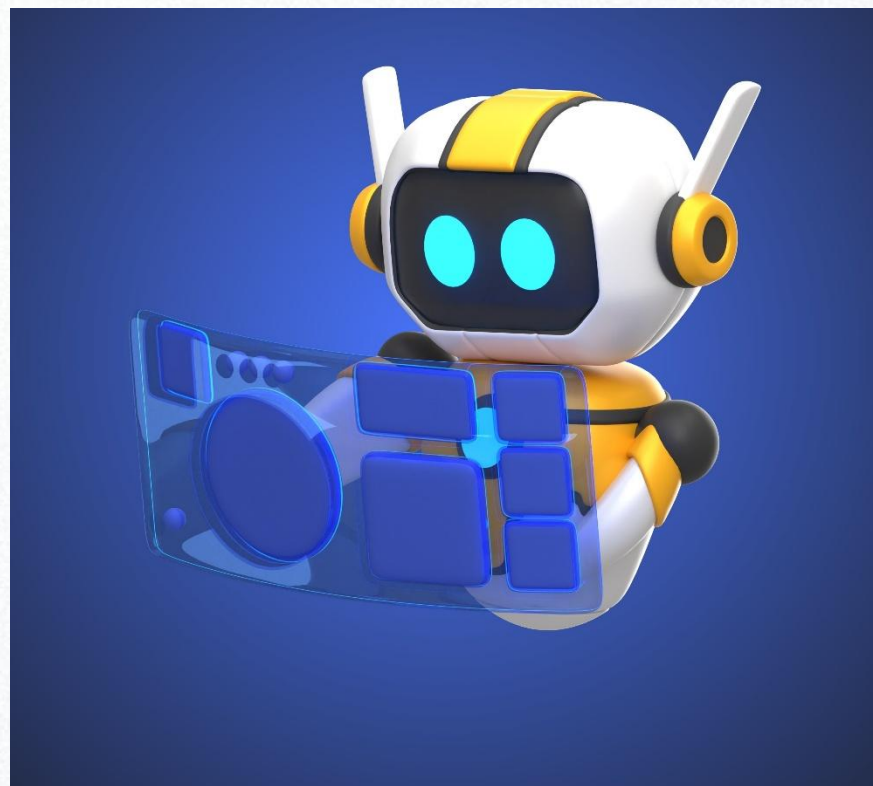
- So‘nggi yillarda dunyoda geologik tahlil sohasida sun’iy intellekt texnologiyalaridan foydalanish bo‘yicha sezilarli yutuqlarga erishilmoqda. Xususan, mineralogik va petrografik tahlil jarayonlarida **Computer Vision** va **Deep Learning** usullarining qo‘llanilishi ilmiy va sanoat amaliyotida katta ahamiyat kasb etmoqda. Bu bo‘yicha qilingan ishlar xalqaro miqyosdagi maqolalarda o‘z aksini topgan. AQSh, Kanada, Avstraliya, Xitoy va Yaponiya kabi geologiya sohasida yetakchi mamlakatlarda mikroskopik tasvirlarni tahlil qilishda **CNN (Convolutional Neural Networks)** modellari asosidagi tadqiqotlar bo‘yicha erishilgan natijalar soha mutaxassislari tomonidan e’tirof etilmoqda.

- Petrologik tadqiqotlarda foydalanadigan **PetroExplorer**, **Petroledge** dastrurlari hamda **Alex Strekeisen**, **Pax-it**, **Petrog**, **FastGAPP** va **Petra** saytlari mavjud. Ulardan diagrammalar tayyorlashda, shlif hamda anshlif rasmlariga fotoishlov berishda foydalanamiz. Ammo, yuqoridagi vositalar barchasi statistik ma’lumotlar asosida ishlaydi hamda shlif-anshlif-ICP integratsiyaga ega emas. Zamonaviy tadqiqotlar esa kompleks tahlillar, xalqaro standartlar asosidagi natijalarni talab qiladi.

- Mavjud ishlanmalarning ko‘pchiligi faqat tadqiqot bosqichida bo‘lib, sanoatga to‘liq integratsiyalashmagan yoki foydalanuvchi uchun qulay interfeysga ega emas. Petrolog AI loyihasi ushbu bo‘shliqni to‘ldirishga qaratilgan bo‘lib, mikroskopik tasvirlar asosida minerallarni aniqlash, geokimyoviy tahlillarni integratsiya qilish va tayyor hisobot yaratishni bir platformada birlashtiradi. Loyiha o‘zbek geologiyasida ilk marotaba bunday sun’iy intellekt yondashuvini keng ko‘lamda joriy etishga intiladi.



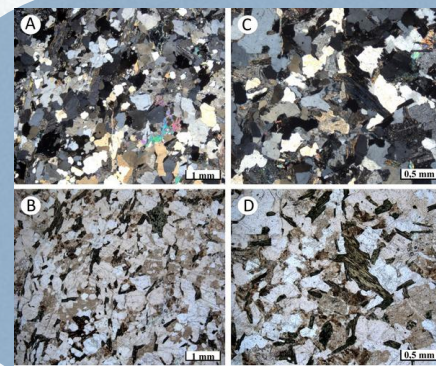




Loyiha O'zbekiston sharoitida ilk bor sun'iy intellekt texnologiyalarini geologiya sohasi bilan integratsiyasiga asoslangan, raqamli laboratoriya tizimini yaratish orqali milliy ilmfan va sanoatning dolzarb ehtiyojlarini qondiradi.

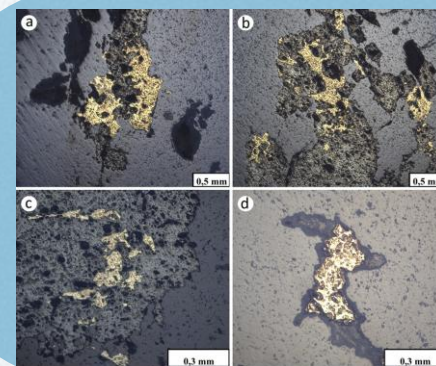
Shu bois **Petrolog AI** — nafaqat innovatsion, balki hozirgi va kelajak ehtiyojlariga javob bera oladigan strategik ahamiyatga ega loyihadir.

# LOYIHANING DOLZARBLIGI



# Shaffof shlif

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 adipiscing elit, sed do eiusmod.



# Sayqallangan shlif

Lorem ipsum dolor sit amet, consectetur  
 adipiscing elit, sed do eiusmod.

| Star ID | LT                | Re                | T <sub>1</sub>    | Na <sup>+</sup> | Mg <sup>+</sup> | Al <sup>+</sup> | P     | Si <sup>+</sup>   | Ca <sup>+</sup>  | Sc                | T <sub>2</sub>    | Li                | Cr                |
|---------|-------------------|-------------------|-------------------|-----------------|-----------------|-----------------|-------|-------------------|------------------|-------------------|-------------------|-------------------|-------------------|
|         | 2                 | 3                 | 4                 | 5               | 6               | 7               | 8     | 9                 | 10               | 11                | 12                | 13                | 14                |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 4244 ± 120      | 3000 ± 110      | 6121 ± 81       | 14200 | 5200 <sup>+</sup> | 710              | 1400 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 2110 ± 100      | 1500 ± 100      | 3100 ± 100      | 14200 | 5200 <sup>+</sup> | 710              | 1400 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 1500 ± 100      | 1000 ± 100      | 2100 ± 100      | 14200 | 5200 <sup>+</sup> | 710              | 1400 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 1000 ± 100      | 500 ± 100       | 1000 ± 100      | 14200 | 5200 <sup>+</sup> | 710              | 1400 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 500 ± 100       | 200 ± 100       | 500 ± 100       | 14200 | 5200 <sup>+</sup> | 710              | 1400 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 14000 ± 1000    | 10000 ± 1000    | 15000 ± 1000    | 400   | 500 <sup>+</sup>  | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 10000 ± 1000    | 5000 ± 1000     | 10000 ± 1000    | 400   | 500 <sup>+</sup>  | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 5000 ± 1000     | 2000 ± 1000     | 5000 ± 1000     | 400   | 500 <sup>+</sup>  | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 2000 ± 1000     | 1000 ± 1000     | 2000 ± 1000     | 400   | 500 <sup>+</sup>  | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 1000 ± 1000     | 500 ± 1000      | 1000 ± 1000     | 400   | 500 <sup>+</sup>  | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 500 ± 1000      | 200 ± 1000      | 500 ± 1000      | 400   | 500 <sup>+</sup>  | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 13300 ± 1000    | 10000 ± 1000    | 15000 ± 1000    | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 10000 ± 1000    | 5000 ± 1000     | 10000 ± 1000    | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 5000 ± 1000     | 2000 ± 1000     | 5000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 2000 ± 1000     | 1000 ± 1000     | 2000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 1000 ± 1000     | 500 ± 1000      | 1000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 500 ± 1000      | 200 ± 1000      | 500 ± 1000      | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 10000 ± 1000    | 5000 ± 1000     | 10000 ± 1000    | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 5000 ± 1000     | 2000 ± 1000     | 5000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 2000 ± 1000     | 1000 ± 1000     | 2000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 1000 ± 1000     | 500 ± 1000      | 1000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 500 ± 1000      | 200 ± 1000      | 500 ± 1000      | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 10000 ± 1000    | 5000 ± 1000     | 10000 ± 1000    | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 5000 ± 1000     | 2000 ± 1000     | 5000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 2000 ± 1000     | 1000 ± 1000     | 2000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 1000 ± 1000     | 500 ± 1000      | 1000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 500 ± 1000      | 200 ± 1000      | 500 ± 1000      | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 10000 ± 1000    | 5000 ± 1000     | 10000 ± 1000    | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 5000 ± 1000     | 2000 ± 1000     | 5000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 2000 ± 1000     | 1000 ± 1000     | 2000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 1000 ± 1000     | 500 ± 1000      | 1000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 500 ± 1000      | 200 ± 1000      | 500 ± 1000      | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 10000 ± 1000    | 5000 ± 1000     | 10000 ± 1000    | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 5000 ± 1000     | 2000 ± 1000     | 5000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 2000 ± 1000     | 1000 ± 1000     | 2000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 1000 ± 1000     | 500 ± 1000      | 1000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 500 ± 1000      | 200 ± 1000      | 500 ± 1000      | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 10000 ± 1000    | 5000 ± 1000     | 10000 ± 1000    | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 5000 ± 1000     | 2000 ± 1000     | 5000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 2000 ± 1000     | 1000 ± 1000     | 2000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 1000 ± 1000     | 500 ± 1000      | 1000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 500 ± 1000      | 200 ± 1000      | 500 ± 1000      | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 10000 ± 1000    | 5000 ± 1000     | 10000 ± 1000    | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 5000 ± 1000     | 2000 ± 1000     | 5000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 2000 ± 1000     | 1000 ± 1000     | 2000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 1000 ± 1000     | 500 ± 1000      | 1000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 500 ± 1000      | 200 ± 1000      | 500 ± 1000      | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 10000 ± 1000    | 5000 ± 1000     | 10000 ± 1000    | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 5000 ± 1000     | 2000 ± 1000     | 5000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 2000 ± 1000     | 1000 ± 1000     | 2000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 1000 ± 1000     | 500 ± 1000      | 1000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
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| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 10000 ± 1000    | 5000 ± 1000     | 10000 ± 1000    | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 5000 ± 1000     | 2000 ± 1000     | 5000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
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| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 1000 ± 1000     | 500 ± 1000      | 1000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
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| K2-12   | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 2000 ± 1000     | 1000 ± 1000     | 2000 ± 1000     | 710   | 1000 <sup>+</sup> | 100 <sup>+</sup> | 1000 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> | 15.0 <sup>+</sup> |
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## ICP MS tahlili

Lorem ipsum dolor sit amet, consectetur  
 adipiscing elit, sed do eiusmod.



# GQI va X/Sh

12 ta Geologik-qidiruv ishlari va 27 ta xo'jalik shartnomalari mavjud.



# Loyihaning amaliy ahamiyati

**Petrolog AI** loyihasi geologik tahlil sohasida ilmiy va ishlab chiqarish jarayonlarini avtomatlashtirish orqali amaliy samaradorlikni sezilarli darajada oshiradi. Loyiha yordamida mikroskop ostida olingan shlif va anshlif tasvirlarini sun'iy intellekt yordamida avtomatik tahlil qilish imkoniyati yaratiladi, bu esa quyidagi muhim afzalliklarni beradi

1

## Vaqt va resurs tejaladi

An'anaviy tarzda bir necha soat yoki kun talab qiladigan tahlillar Petro AI orqali bir necha daqiqada bajariladi

2

## Aniqlik oshadi

Sun'iy intellekt asosidagi model insoniy subyektivlikni kamaytiradi va tahlil natijalarining izchilligini ta'minlaydi.

3

## Xalqaro tajribalarga asoslangan usullar bilan ishlash

Sun'iy intellekt ma'lumotlar bazasidagi xalqaro tajribalarga tayangan pettrografik xulosalarni olish imkoniyatiga ega bo'ladi

4

## Ta'lim va ilmiy izlanishlar uchun foydali

Talabalar va tadqiqotchilar murakkab tahlillarni tez va aniq amalga oshirish orqali o'quv va ilmiy ishlarini sifatli bajarish imkoniga ega bo'ladilar

5

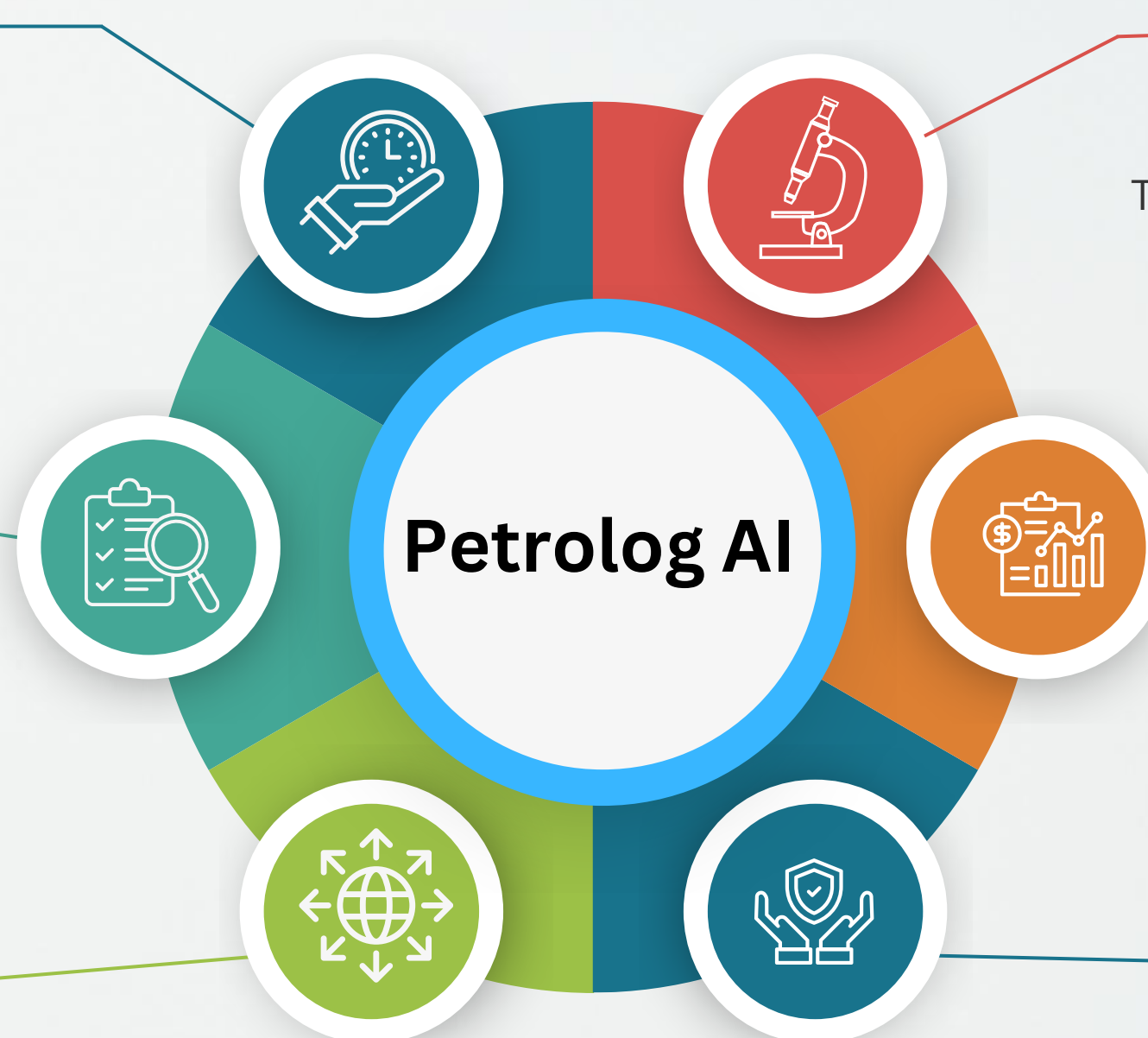
## Sanoatga moslashuvchanlik

Konlar, qazilma boyliklar, qurilish materiallari yoki burg'ulash ishlarida minerallarni tez aniqlash orqali ishlab chiqarish qarorlarini tezlashtirish mumkin bo'ladi.

6

## Zamonaviy yondoshuv

Petrolog AI platformasi O'zbekiston geologiya sohasini raqamlashtirishga xizmat qiladi, ilm-fan va texnologiyani uzviy bog'lash orqali milliy geologik tahlil infratuzilmasining zamonaviylashishiga hissa qo'shadi



## Loyihadan ko'zda tutilgan maqsadlar

01

Raqamli  
laboratoriya tizimi  
shakllantiriladi

02

Mineralogik-  
petrografik tahlillar  
avtomatlashtiriladi

03

Geologik ma'lumotlar  
bilan ishlashda vaqt va  
resurslar tejaladi

04

Ilmiy va sanoat ehtiyojlar  
uchun zamonaviy vosita  
yaratiladi

## Loyihaning maqsad va vazifalari

**Petrolog AI platformasining asosiy maqsadi**  
— mikroskop orqali olingan shlif va anshlif  
tasvirlarini hamda kimyoviy analiz  
natijalarini sun'iy intellekt yordamida tahlil  
qilish orqali tog' jinslarining moddiy  
tarkibini aniqlash, ularning tavsiflash va bu  
orqali geologik jihatdan tahlil qilish

## Loyihaning ustuvor vazifalari

05

Shlif-anshlif rasmlari va  
qo'shimcha ma'lumotlar  
asosida SI yordamida tahlil  
qiluvchi modeli yaratish

06

Turli xil minerallarni tasvir  
asosida aniqlovchi  
algoritmni ishlab chiqish

07

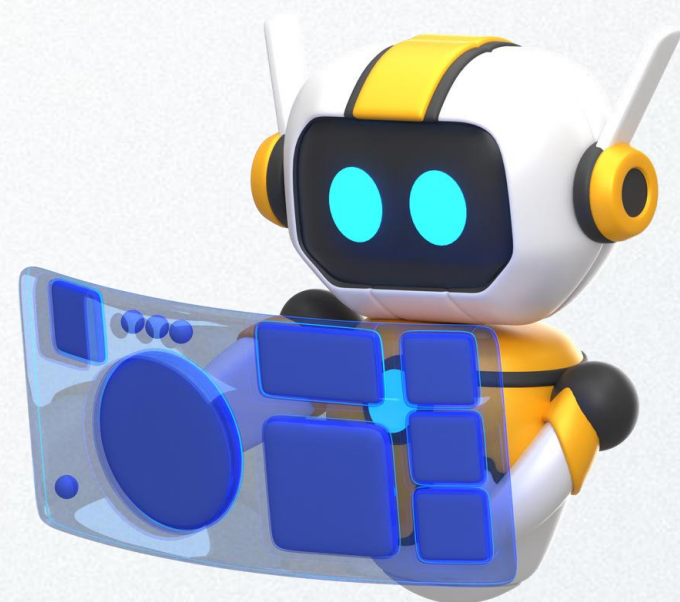
Oddiy va qulay  
foydalanuvchi  
interfeysini yaratish

08

Tahlil natijalari asosida  
ilmiy va amaliy tavsiyalar  
ishlab chiqish



# PETROLOG AI interfeysi



## AI TAHLILI TANLOVLARI



Shlif tahlili



Anshlif tahlili



ICP tahlili



Kompleks tahlil

Shaffof shlif rasmini yuklash



Izoh: Format: TIFF, PNG, JPG, 10 mb dan oshmagan, o'lcham aniqlash uchun belgilarga ega bo'lishi kerak\*

Sayqallangan shlif rasmini yuklash



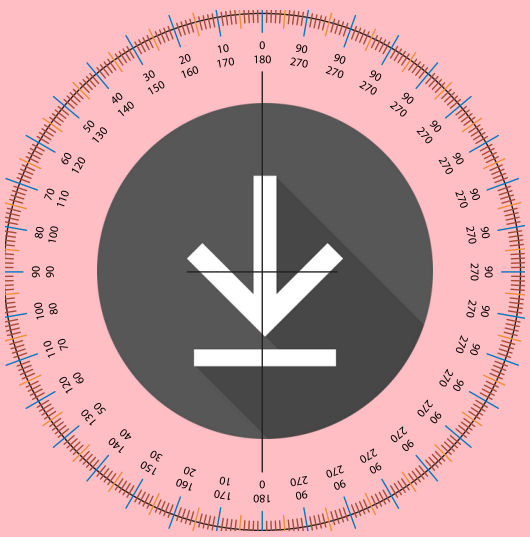
Izoh: Format: TIFF, PNG, JPG, 10 mb dan oshmagan, o'lcham aniqlash uchun belgilarga ega bo'lishi kerak\*

ICP MS natijalarini yuklash



Izoh: Format: CSV, XLSX, 10 mb dan oshmagan, excel, word yoki pdf formatda bo'lishi kerak\*

Sayqallangan shlif 360° videosini yuklash



Izoh: Format: MP4, MOV, 20 mb dan oshmagan, 360° ga aylantirilgan bo'lishi kerak\*



AI tahlil







Asosiy

Tahlil

Natijalar

Bog'lanish

Yuklanish

Shlif tasviri

Bajarildi

Anshlif tasviri

Kutilmoqda

ICP fayl

Format xato

Shlif videosi

Bajarildi

AI tahlili tanlovlari

Video yo'riqnoma



Shlif tahlili



Anshlif tahlili



ICP tahlili



Kompleks tahlil

Analitik tahlil ko'rsatkichi

75%

Namuna ma'lumotlari

| Parametr           | Qiymat                |
|--------------------|-----------------------|
| Namuna kodi        | ZZ - 1                |
| Tog' jinsi turi    | Granodiorit           |
| Mikroskop modeli   | Nikon ECLIPSE Iv 1    |
| Ob'ektiv kattaligi | 20x (shlif), 40x (an) |

AI tahlil  
natijalari

| Mineral             | Ulushi (%) | Aniqlik (conf.) | Pleox.   | Optik belgi                   | Izoh                              |
|---------------------|------------|-----------------|----------|-------------------------------|-----------------------------------|
| Kvars               | 48.6       | 0.97            | Yo'q     | To'liq aniqlik                | Nisbatan yedirilgan               |
| Plagioklaz          | 23.4       | 0.91            | Kam      | Zonal tuzilish                | Seritsitlashgan                   |
| Biotit              | 16.1       | 0.93            | Kuchli   | Qizg'ish-jigarrang - sarg'ish | O'zgargan                         |
| Amfibol             | 7.8        | 0.84            | O'rtacha | Yashil-ko'k pleoxroizm        | Uzun kristallarga ega             |
| Ma'danli minerallar | 4.1        | 0.79            | —        | Metallik yaltirash            | Mayda tarqoq donachalardan iborat |

AI tahlilidan generatsiya qilingan ilmiy tavsif

Namuna kvarts (48.6%), plagioklaz (23.4%) va biotit (16.1%)dan tashkil topgan bo'lib, amfibol va magnetit ma'danli minerallari qo'shimcha komponent sifatida ishtirok etadi.

Optik belgilar oraliq kislotali intruziv jins — granodiorit tarkibiga mos keladi. Plagioklazda serikitleanish, biotitda gidrotit almashinuvi kuzatiladi. ICP natijalari SiO<sub>2</sub> 62.8% bo'lishini ko'rsatadi, bu oraliq kislotali diapazonga mosdir. Fe/Mg nisbati 1.86 — metamorfik o'zgarishning o'rtacha bosqichini bildiradi.

Umuman, AI-integratsiya natijasi bo'yicha namuna granodiorit intruziv kompleksiga mansub.

Xulosa

- AI klassifikatsiyasi: Granodiorit
- Xalqaro moslik: IUGS (Streckeisen, 1973) diagrammasi bilan mos
- AI ishonchliligi: 88.8%
- Ekspert tasdiqlash: Ha
- Model versiyasi: v0.4 (FusionNet)

Yuklash ISO/ASTM



Yuklash doc.





# ISO/ASTM LABORATORIYA HISOBOT FORMATI

Hujjat kodi: GGI 05/2025

Standart asos: ISO/IEC 17025:2017, ASTM D2798-20, ISO 14577-1:2015

Hisobot turi: Kompleks petrografik va kimyoviy tahlil

## 1. Umumiy ma'lumotlar

| Parametr             | Ma'lumot                                          |
|----------------------|---------------------------------------------------|
| Laboratoriya         | Petrograf AI — AI asosida geologik tahlil markazi |
| Hisobot raqami       | 05/2025 loyiha                                    |
| AI modeli            | Petrolog AI v0.4<br>(PyTorch, multimodal fusion)  |
| Ekspert tasdiqlovchi | _____                                             |

## 2. Namuna ma'lumotlari

| Parametr           | Qiyamat                    |
|--------------------|----------------------------|
| Namuna kodi        | ZZ - 1                     |
| Tog' jinsi turi    | Granodiorit                |
| Mikroskop modeli   | Nikon ECLIPSE lv 100       |
| Ob'ektiv kattaligi | 20× (shlif), 40× (anshlif) |

## 3. AI tahlil natijalari

| Mineral             | Ulushi (%) | Aniqlik (conf.) | Pleoxroizm | Optik belgi                   | Izoh                              |
|---------------------|------------|-----------------|------------|-------------------------------|-----------------------------------|
| Kvars               | 48.6       | 0.97            | Yo'q       | To'liq aniqlik                | Nisbatan yedirilgan               |
| Plagioklaz          | 23.4       | 0.91            | Kam        | Zonal tuzilish                | Seritsitlashgan                   |
| Biotit              | 16.1       | 0.93            | Kuchli     | Qizg'ish-jigarrang - sarg'ish | O'zgargan                         |
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| Ma'danli minerallar | 4.1        | 0.79            | —          | Metallik yaltirash            | Mayda tarqoq donachalardan iborat |

## 4. AI tahlilidan generatsiya qilingan ilmiy tavsif

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## Xulosa

- AI klassifikatsiyasi: **Granodiorit**
- Xalqaro moslik: IUGS (Streckeisen, 1973) diagrammasi bilan mos
- AI ishonchliligi: 88.8%
- Ekspert tasdiqlash: **Ha**
- Model versiyasi: v0.4 (FusionNet)

## Lavozim

AI tizimi operatori

Ekspert geolog

Laboratoriya rahbari

## F.I.Sh.

J. Xudoyberdiyev

G. Xolmurodov

N. Usmonova

## Imzo

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

# YAXLIT ILMIY-ANALITIK TEKST SHAKLIDAGI TAHLIL

## Tahlil natijasi:

AI yordamida tahlil qilingan SHLIF-2025-021 namunasi optik va kimyoviy ma'lumotlarning kompleks integratsiyasiga asoslanadi. Shlif tasvirida kvarts (taxminan 49%) asosiy komponent sifatida qayd etilgan bo'lib, u past relyefli, aniq interferensiya ranglariga ega, pleoxroizm belgilari kuzatilmagan. Plagioklaz (23%) seritsitlanish belgilari bilan aniqlanadi; u zonal tuzilishga ega bo'lib, qisman Ca–Na almashinuvi ko'rinadi. Biotit (16%) kuchli pleoxroizm (qizg'ish-jigarrang → sarg'ish) bilan ajralib turadi va qisman gidrotitga aylangan. Amfibol (8%) yashil-ko'k rangli pleoxroizmga ega, uzun kristalli shaklda joylashgan. Ruda minerallar (4%) pirit va magnetit ko'rinishida reflektiv yuzalarda aniqlangan.

ICP-MS tahlilida SiO<sub>2</sub> miqdori 62.8% bo'lib, bu tog' jinsni nordon tarkibli ekanini ko'rsatadi. Al<sub>2</sub>O<sub>3</sub> (16.3%), Fe<sub>2</sub>O<sub>3</sub> (5.2%) va MgO (2.8%) nisbati jinsning biotit–amfibol bilan boyigan tarkibini ko'rsatadi. Fe/Mg nisbati 1.86 bo'lib, bu metamorfik o'zgarishning o'rtacha bosqichini bildiradi. K/Na nisbati 0.83 bo'lishi K-feldspat va plagioklaz orasidagi muvozanatni ko'rsatadi.

AI modeli tomonidan hosil qilingan integratsiyalangan fazoviy xususiyatlar (segmentatsiya maskalari, rang tahlili, kimyoviy vektorlar) birgalikda interpretatsiya qilinib, namunaning tarkibi **granodiorit intruziv** jinslariga mosligi aniqlangan. Shlifda kuzatilgan biotit–amfibol birikmalari va ICP natijasidagi Fe/Mg nisbati metamorfik qayta kristallannish belgilari bilan mos keladi.

Namuna **IUGS Streckeisen (1973)** diagrammasida granodiorit sohasi ichiga tushadi. AI modelining o'rtacha ishonchlilik darajasi 0.888 bo'lib, ekspert bahosi bilan moslik koeffitsienti 0.82 ni tashkil etdi.

**Xulosa:** SHLIF-2025-021 — kvarts–plagioklaz–biotit–amfibol tarkibli, nordon intruziv tog' jinsi. Petrografik va kimyoviy tahlillar kompleks xolatda ushbu namuna granodiorit ekanligini ko'rsatadi.